

The “Difficulties” in Fermat’s Original Discourse on the Indecomposability of Powers Greater Than a Square: A Retrospect

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Abstract

We present a reconstruction of Pierre Fermat’s possible manuscript reasoning based on the two-power decomposition constraint, on the coefficient symmetry of the odd part of the binomial expansion, and on a logarithmic compatibility reading of the two residual scales.

Starting from a hypothetical positive natural solution of

$$x^n + y^n = z^n, \quad n > 2,$$

we use the symmetric parametrization

$$z = m^n + p^n, \quad x = m^n - p^n,$$

where m and p are first treated as positive real parameters. This gives

$$y^n = (m^n + p^n)^n - (m^n - p^n)^n.$$

The odd part of this binomial difference admits two natural real normalizations. The first extracts the linear binomial factor $n = C_n^1$:

$$y^n = 2n l^n.$$

The second extracts the total odd binomial coefficient mass. By the symmetry of Pascal’s triangle,

$$\sum_{r \text{ odd}} C_n^r = 2^{n-1},$$

and hence the second normalization has the form

$$y^n = 2^n q^n.$$

Here l and q are positive real residual factors. The symbol q denotes a residual factor and must not be confused with a modulus in modular arithmetic.

Comparing the two representations gives the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Since $l > 0$ and $q > 0$, this relation can equivalently be written in logarithmic form as

$$\Delta_n := \ln \frac{l}{q} = \frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

The quantity Δ_n measures the logarithmic imbalance between the two residual factors. It may be interpreted heuristically as an entropy-type gap between the two normalizations of the same odd binomial core, but in the strict mathematical argument it is only a logarithmic reformulation of the exact algebraic comparison.

Thus, within the two real normalizations of the same odd binomial core, the residual-scale equality

$$l^n = q^n$$

is equivalent to the zero-gap condition

$$\Delta_n = 0.$$

Both conditions are equivalent to the equality between the linear coefficient n and the total odd coefficient mass 2^{n-1} :

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1}.$$

The latter equation has exactly two positive integer solutions,

$$n \in \{1, 2\}.$$

Moreover, for every integer $n > 2$, one has

$$2^{n-1} > n,$$

and therefore

$$\Delta_n > 0.$$

Consequently, the binomial coefficient symmetry gives a logarithmic compatibility obstruction: the zero-gap condition, equivalently the equality of the two residual scales, is possible only in the linear and quadratic cases, and necessarily fails for every exponent $n > 2$.

Therefore, if the internal two-power symmetry of a hypothetical positive integer solution is required to preserve zero logarithmic gap between the two residual factors, then every hypothetical solution with $n > 2$ leads to a contradiction.

The decomposition premise is kept as an explicit structural assumption over $\mathbb{N}$. The real parametrization is used as an algebraic reconstruction, while the integer specialization, parity conditions, and modular caveat are treated separately. In particular, modular congruences are not identified with integer equalities. The accompanying Coq formalization verifies the final exponential-linear obstruction

$$n = 2^{n-1} \implies n \in \{1, 2\},$$

and related bridge lemmas. In the present logarithmic formulation, the role of binomial symmetry is sharpened further: it explains why the logarithmic gap vanishes in the quadratic case and why it becomes strictly positive for every exponent higher than the square. However, unless one proves independently that a hypothetical positive integer solution must force the zero-gap condition, the argument remains a conditional reconstruction rather than an unconditional proof of Fermat's Last Theorem.

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1 Introduction

The present work is the result of an attempted reconstruction of Fermat's original discourse along with an explanation of why he might have not written it down. The author had performed it within a two-time period of time—between 1990 and 1993—trying to prove the theorem and the final revision in 2017-2026.¹ When completed, the reconstruction appeared to the author as a possible Fermat-era line of reasoning, since its algebraic language relies only on elementary operations, binomial expansion, and logarithmic comparison. In its present form, however, it is not presented as an unconditional proof, but as a conditional reconstruction whose decisive point is the zero-gap compatibility requirement. Not to overburden this text with details of a real historical study, let us briefly recall the history of the conjecture. Around 1637, Fermat wrote his Last Theorem in the margin of his copy of the *Arithmetica* next to Diophantus' sum-of-squares problem [7]:

Cubum autem in duos cubos, aut quadratoquadratum in duos quadratoquadra-tos & generaliter nullam in infinitum ultra quadratum potestatem in duos eiusdem nominis fas est dividere cuius rei demonstrationem mirabilem sane detexi. Hanc marginis exiguitas non caperet.

Attempts to prove this conjecture employed a diverse range of methods. Early attempts, including Fermat's own proof for the case $n = 4$, utilized the **method of infinite descent**. A significant

¹ The work was mainly performed on the scholarship of the NRNU MEPhI student in 1990 – 93, final revision in 2017 – 2026.

step forward was Leonhard Euler's proof for the case $n = 3$, which involved the use of **Gaussian integers** [13]. Subsequent efforts included approaches based on the work of Sophie Germain and other techniques, before Andrew Wiles finally presented a complete proof in 1995, drawing upon deep results from **the theory of modular forms and elliptic curves** [1–6].

Modern methods, such as the theory of modular forms, deal with **the transformation properties of specific curves over particular types of spaces** (e.g., rational numbers), highlighting the **stability of elliptic curves with respect to modular transformations**. While the contemporary formalizations of algebraic curves, spaces, transformations, and groups used in these methods were not present in Fermat's time, mathematicians of that era possessed their own approaches and intuitive understandings for studying the properties of natural numbers (and primes). (More on this can be found in the Conclusions section of this paper.)

Theorem 1 (Logarithmic coefficient-symmetry obstruction for the two-power decomposition). *Assume that any hypothetical positive natural solution of the Fermat equation*

$$x^n + y^n = z^n$$

with $n > 2$ admits the real symmetric parametrization

$$z = m^n + p^n, \quad x = m^n - p^n,$$

with positive real parameters $m, p > 0$. Then

$$y^n = (m^n + p^n)^n - (m^n - p^n)^n.$$

Let the odd binomial core of this difference be represented in two natural ways by positive real residual factors $l, q > 0$:

$$(m^n + p^n)^n - (m^n - p^n)^n = 2n l^n,$$

and

$$(m^n + p^n)^n - (m^n - p^n)^n = 2^n q^n.$$

Here l is the residual factor obtained after extracting the linear binomial factor $n = C_n^1$, while q is the residual factor obtained after extracting the total odd binomial coefficient mass 2^{n-1} . In this theorem, q is a positive real residual factor and must not be confused with a modulus in modular arithmetic.

Then the comparison of these two natural normalizations gives the exact compatibility relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Equivalently, since $l > 0$ and $q > 0$, it gives the logarithmic gap

$$\Delta_n := \ln \frac{l}{q} = \frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

Consequently, the residual-scale equality

$$l^n = q^n$$

is equivalent to the zero-gap condition

$$\Delta_n = 0,$$

and both are equivalent to

$$n = 2^{n-1}.$$

Since this equation has only the positive integer solutions $n = 1$ and $n = 2$, the zero-gap condition is incompatible with every exponent $n > 2$.

Therefore, within the zero-gap compatibility framework, if the internal two-power symmetry of a hypothetical positive integer solution is required to preserve zero logarithmic gap between the two residual factors, equivalently to preserve the equality of the two residual scales, then no such compatible positive natural solution can exist for any exponent $n > 2$.

Proof sketch. Given a putative positive natural solution at a fixed exponent $n > 2$, the two-power decomposition gives the binomial identity

$$y^n = (m^n + p^n)^n - (m^n - p^n)^n.$$

The odd part of the binomial expansion admits two real normalizations:

$$y^n = 2n l^n$$

and

$$y^n = 2^n q^n.$$

The first normalization extracts the linear coefficient $n = C_n^1$. The second normalization extracts the full odd coefficient mass. By the symmetry of Pascal's triangle,

$$\sum_{r \text{ odd}} C_n^r = 2^{n-1}.$$

Thus the two extracted coefficient masses are n and 2^{n-1} .

Equating the two expressions for y^n gives

$$2n l^n = 2^n q^n.$$

Since $2n > 0$, division by $2n$ gives the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

This relation is not an additional hypothesis; it follows from the two normalizations of the same odd binomial core.

Since $l > 0$ and $q > 0$, we may take logarithms. From

$$l^n = q^n \frac{2^{n-1}}{n}$$

we obtain

$$n \ln l = n \ln q + (n-1) \ln 2 - \ln n.$$

After division by n , this becomes

$$\ln l - \ln q = \ln 2 - \frac{\ln(2n)}{n}.$$

Equivalently,

$$\ln \frac{l}{q} = \frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

Define the logarithmic gap by

$$\Delta_n := \ln \frac{l}{q} = \frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

This quantity measures the logarithmic imbalance between the two residual factors obtained from the two normalizations of the same odd binomial core. It may be interpreted heuristically as an entropy-type gap, but in the strict mathematical argument it is only a logarithmic reformulation of the exact algebraic relation above.

Now suppose that the two residual scales are required to coincide:

$$l^n = q^n.$$

Since $l > 0$ and $q > 0$, this is equivalent to

$$l = q,$$

and hence to

$$\Delta_n = 0.$$

Using the formula for Δ_n , the zero-gap condition gives

$$\frac{1}{n} \ln \frac{2^{n-1}}{n} = 0.$$

Therefore,

$$\ln \frac{2^{n-1}}{n} = 0,$$

and hence

$$\frac{2^{n-1}}{n} = 1.$$

Thus

$$n = 2^{n-1}.$$

Conversely, if

$$n = 2^{n-1},$$

then

$$\frac{2^{n-1}}{n} = 1,$$

so

$$\Delta_n = \frac{1}{n} \ln 1 = 0.$$

Therefore $l = q$, and since $l, q > 0$, this is equivalent to

$$l^n = q^n.$$

Thus, under the two natural real normalizations of the same odd binomial core, we have the exact equivalence

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1}.$$

The equation

$$n = 2^{n-1}$$

has exactly two positive integer solutions:

$$n \in \{1, 2\}.$$

Indeed, it holds for $n = 1$ and $n = 2$, while for every integer $n > 2$ the exponential term 2^{n-1} strictly dominates the linear term n :

$$2^{n-1} > n.$$

Therefore

$$\frac{2^{n-1}}{n} > 1$$

for every $n > 2$, and consequently

$$\Delta_n = \frac{1}{n} \ln \frac{2^{n-1}}{n} > 0.$$

Hence the zero-gap condition is impossible for every exponent higher than the square.

Thus the binomial coefficient symmetry does not merely motivate the second normalization. It gives a precise logarithmic compatibility obstruction: the condition

$$\Delta_n = 0$$

or, equivalently,

$$l^n = q^n,$$

is compatible with the two binomial normalizations only in the cases

$$n = 1 \quad \text{or} \quad n = 2.$$

Consequently, if the internal two-power symmetry of a hypothetical positive integer solution is required to preserve zero logarithmic gap between the two residual factors, then a putative solution with $n > 2$ leads to a contradiction.

The elementary growth comparison

$$n = 2^{n-1} \implies n \in \{1, 2\}$$

is the same final exponential-linear obstruction that is machine-checkable in the accompanying Coq file currently named `GlobalNormalization.v`.

In the present logarithmic formulation, the role of the binomial symmetry is sharpened further: it explains why the logarithmic gap vanishes in the quadratic case and why it is strictly positive for every exponent $n > 2$. This logarithmic gap may be viewed as an entropy-type diagnostic of the separation between the two residual factors. However, unless one proves independently that a hypothetical positive integer solution must force the zero-gap condition, the argument remains a conditional reconstruction rather than an unconditional proof of Fermat's Last Theorem.

2 Conditional Algebraic Reconstruction

The classical statement of Fermat's Last Theorem is as follows: no power greater than the square can be decomposed into two powers of the same exponent. Equivalently, the equation

$$x^n + y^n = z^n$$

has no positive integer solutions for $n > 2$.

In the present paper, this statement is not reproved unconditionally. What follows is a conditional algebraic reconstruction based on the zero-gap compatibility requirement.

Therefore, first

1). Let's write down the theorem

$$x^n + y^n = z^n \tag{2.1}$$

2). Let's rewrite (2.1)

$$z^n - x^n = y^n \tag{2.2}$$

3). Let's perform the transformation of (2.2) to

$$(m^n + p^n)^n - (m^n - p^n)^n = y^n, \tag{2.3}$$

where $n \in \mathbb{N}$, m, p are arbitrary numbers (not necessarily integers; signs arbitrary), and $z, x \in \mathbb{N}$ satisfy

$$\begin{cases} m^n + p^n = z, \\ m^n - p^n = x. \end{cases} \tag{2.4}$$

Raising both identities to the n -th power yields

$$\begin{cases} (m^n + p^n)^n = z^n, \\ (m^n - p^n)^n = x^n. \end{cases} \tag{2.5}$$

Thus (2.4) implies (2.5).

Solving (2.4). Adding and subtracting the equations gives

$$2m^n = z + x, \quad 2p^n = z - x,$$

so formally

$$m = \sqrt[n]{\frac{z+x}{2}}, \quad p = \sqrt[n]{\frac{z-x}{2}}.$$

Domain note. Over $\mathbb{R}$: for odd n the roots are unique; for even n we need $(z \pm x)/2 \geq 0$ and there is a sign choice $m = \pm((z+x)/2)^{1/n}$, $p = \pm((z-x)/2)^{1/n}$. Over $\mathbb{C}$: there are n branches for the n -th root; once a branch is fixed, the reconstruction is consistent.

Equivalence of (2.4)–(2.5). Forward (2.4) $\Rightarrow$ (2.5). Immediate by raising to the n -th power.

Reverse (2.5) $\Rightarrow$ (2.4). Assume $z, x \in \mathbb{N}$ and

$$z^n = (m^n + p^n)^n, \quad x^n = (m^n - p^n)^n.$$

Taking n -th roots in the same domain as $m^n \pm p^n$ (principal root over $\mathbb{R}_{\geq 0}$, or a fixed branch over $\mathbb{C}$) gives

$$z = m^n + p^n, \quad x = m^n - p^n,$$

i.e. (2.4). For even n one fixes consistent signs as above. Hence, (2.4) and (2.5) are equivalent in the chosen number domain once the root/branch convention is fixed.

Proposition 1 (Integer case: necessary & sufficient conditions). *If, in addition, one requires $m, p \in \mathbb{Z}$, then necessarily*

$$z \pm x \text{ are even}, \quad \frac{z+x}{2} = m^n, \quad \frac{z-x}{2} = p^n.$$

Conversely, if $z \pm x$ are even and both halves are perfect n -th powers in $\mathbb{Z}$, then the reconstructed m, p are integers (up to signs for even n).

Remark 1. *This separates the general real/complex reconstruction (no parity obstruction) from the integer reconstruction, where parity and perfect-power constraints are essential.*

Example (real domain). For $n = 2$, $m = 3$, $p = 2$:

$$z = 3^2 + 2^2 = 13, \quad x = 3^2 - 2^2 = 5,$$

hence

$$z^2 = 169, \quad x^2 = 25,$$

and reversing via $(z \pm x)/2$ returns $m = 3$, $p = 2$.

Counterexample (integer case). For $n = 3$, $z = 2$, $x = 1$ we have $(z+x)/2 = 3/2$, which is not a perfect cube in $\mathbb{Z}$; hence no integer m, p satisfy (2.4), although real m, p exist.

Modular remark. The present work does not construct a proof by means of modular arithmetic and does not claim the absence of solutions to the corresponding congruences over finite fields. The scheme considered here concerns equalities over the natural, integer, rational, and real numbers, not equations over a finite field $\mathbb{F}_q$.

In particular, from an integer equality

$$x^n + y^n = z^n$$

one indeed obtains the congruence modulo any q :

$$x^n + y^n \equiv z^n \pmod{q}.$$

However, the converse implication is not valid in general: from the congruence

$$x^n + y^n \equiv z^n \pmod{q}$$

one cannot infer the equality

$$x^n + y^n = z^n$$

over the integers.

In other words, a modular congruence only means that there exists an integer t such that

$$z^n = x^n + y^n + qt.$$

An integer solution of Fermat's equation corresponds to the special case

$$t = 0.$$

Therefore, the existence of solutions to the congruences

$$x^n + y^n \equiv z^n \pmod{q}$$

for $n > 2$ is not a counterexample to Fermat's Last Theorem and does not disprove the real parametrization used in the present work.

If one specifically passes to a modulus q , then the parameters must be specified separately at the modular level. For example, instead of the real equalities

$$z = m^n + p^n, \quad x = m^n - p^n$$

one should separately fix the residues

$$m^n \equiv A \pmod{q}, \quad p^n \equiv B \pmod{q},$$

after which the corresponding modular relations take the form

$$z \equiv A + B \pmod{q}, \quad x \equiv A - B \pmod{q}.$$

Thus, the real parametrization and modular arithmetic belong to different levels of description. In this article, the first level is used: equalities over $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$, rather than a separate theory of solutions over finite fields.

- 4). Let's ask ourselves the question: what is the number y ? Is it positive integer or not? If it is not positive integer, then under what conditions will it be natural number? Does its naturalness depend on the degree of n ?

From (2.3) we have the difference:

$$y^n = z^n - x^n = (m^n + p^n)^n - (m^n - p^n)^n = \quad (2.6)$$

that can be expanded or decomposed into a sum according to Newton's binomial [8, 9]:

$$\begin{aligned} &= [(m^n)^n + C_n^1(m^n)^{n-1}p^n + C_n^2(m^n)^{n-2}(p^n)^2 + \dots + C_n^{n-1}m^n(p^n)^{n-1} + (p^n)^n] - \\ &- [(m^n)^n - C_n^1(m^n)^{n-1}p^n + C_n^2(m^n)^{n-2}(p^n)^2 \pm \dots \pm C_n^{n-1}m^n(p^n)^{n-1} \pm (p^n)^n] = \\ &= 2C_n^1(m^n)^{n-1}p^n + 2C_n^3(m^n)^{n-3}(p^n)^3 + \dots + 2C_n^k(m^n)^{n-k}(p^n)^k + \{2C_n^n(p^n)^n\} = \\ &= 2 \sum_{i=0}^k C_n^{(2i+1)}(m^n)^{n-(2i+1)}(p^n)^{(2i+1)} \end{aligned}$$

with $k = (n - 1)/2$ if n is odd and $k = (n - 2)/2$ if n is even.

Rewrite then (2.6) as

$$\begin{aligned} &z^n = x^n + y^n, \\ \text{where } x, y, z \text{ are } &\begin{cases} z = m^n + p^n \\ x = m^n - p^n \\ y = \sqrt[n]{2} \left[\sum_{i=0}^k C_n^{(2i+1)}(m^n)^{n-(2i+1)}(p^n)^{(2i+1)} \right]^{1/n} \end{cases} \end{aligned}$$

with $k = (n - 1)/2$ if n is odd and $k = (n - 2)/2$ if n is even;

we know that in common case for $n = 2$ we have

$$\text{where } x, y, z \text{ are given by } \begin{cases} z = r \cdot (m^2 + p^2), \\ x = r \cdot (m^2 - p^2), \\ y = r \cdot 2mp. \end{cases} \quad (2.7)$$

but in our case, we omit the factor r – is some positive ($r > 0$) integer constant since it is reduced in our calculations. Therefore, we made the transformation (2.3) in order to take into account this case ($n=2$), known since the time of Pythagoras, since the general case should include a particular solution as a subset.

5). scrutinise now the y :

$$y = \sqrt[n]{2} \left[\sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{(2i+1)} \right]^{1/n}. \quad (2.8)$$

In order for the y to maybe a positive integer, $\sqrt[n]{2}$ must leave, since for $n > 1$ $\sqrt[n]{2}$ is an irrational number (see Appendix A). It is thus necessary that the expression

$$\left[\sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{(2i+1)} \right]^{1/n} \quad (2.9)$$

contain some common factor that destroys the radical expression $\sqrt[n]{2}$, let's find out what it is. Otherwise, y is not a positive integer due to the presence of $\sqrt[n]{2}$. Consider now a natural real factorization of this odd binomial core by extracting the linear binomial coefficient $n = C_n^1$:

$$\begin{aligned} & \left[\sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{(2i+1)} \right] = \\ &= C_n^1 (m^n)^{n-1} p^n + C_n^3 (m^n)^{n-3} (p^n)^3 + \dots + C_n^k (m^n)^{n-k} (p^n)^k + \{C_n^n (p^n)^n\} = \\ &= n(m^n)^{n-1} p^n + \frac{n(n-1)(n-2)}{3!} (m^n)^{n-3} (p^n)^3 + \dots + \\ &+ \frac{n(n-1)\dots(n-k+1)}{k!} (m^n)^{n-k} (p^n)^k + \left\{ \frac{n(n-1)\dots 2 \cdot 1}{n!} (p^n)^n \right\} = \\ &= n \cdot \left[(m^n)^{n-1} p^n + \frac{(n-1)(n-2)}{3!} (m^n)^{n-3} (p^n)^3 + \dots + \right. \\ &\left. + \frac{(n-1)\dots(n-k+1)}{k!} (m^n)^{n-k} (p^n)^k + \left\{ \frac{(n-1)\dots 2 \cdot 1}{n!} (p^n)^n \right\} \right] = n \cdot l^n \end{aligned} \quad (2.10)$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even,

Since each term in the expression can be written with a common factor n , the whole sum can be factored over the real domain in the form

$$\sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{2i+1} = n l^n.$$

Here $l = l(n, m, p) > 0$ is the positive real residual factor defined by this factorization:

$$l^n = \frac{1}{n} \sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{2i+1}.$$

Thus, this step should be understood as a real factorization of the odd binomial core after extracting the coefficient n . It is not being used as a claim of integer divisibility.

To show the uniqueness of the decomposition of expression (2.10), we can use the following considerations:

(a) Degree n is a fixed positive integer.

- (b) Exponentiation is an unambiguous operation; for any number, there is a unique value for the degree.
- (c) Addition and subtraction of real numbers are commutative operations, and the result does not depend on the order of actions.

Based on these properties, it can be argued that for fixed m , p and n there is a single result of calculating the expression (2.10). Rearranging the members will not affect the final answer.

We can see from (2.8)–(2.10) that l is a positive real residual factor depending on the fixed values of m , p , and n . At this stage no further arithmetic property of l is assumed.

$$y = \sqrt[n]{2 \cdot n \cdot l}. \quad (2.11)$$

As a result, from (2.6)–(2.11) we get

$$z^n - x^n = (m^n + p^n)^n - (m^n - p^n)^n = y^n. \quad (2.12)$$

Substituting (2.11) into (2.12), we obtain

$$(m^n + p^n)^n - (m^n - p^n)^n = 2n l^n. \quad (2.13)$$

The second symmetric scale and the coefficient-symmetry obstruction.

Let us denote the odd binomial core by

$$S_n(m, p) := \sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{2i+1},$$

where $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even. Then the binomial expansion gives

$$(m^n + p^n)^n - (m^n - p^n)^n = 2S_n(m, p).$$

From the previous real factorization of the odd binomial core we have

$$S_n(m, p) = n l^n,$$

where $l > 0$ is a real residual factor. Hence

$$(m^n + p^n)^n - (m^n - p^n)^n = 2n l^n. \quad (2.13a)$$

On the other hand, by the symmetry of Pascal's triangle,

$$C_n^r = C_n^{n-r},$$

the total mass of the odd binomial coefficients is equal to one half of the full binomial mass:

$$\sum_{i=0}^k C_n^{(2i+1)} = 2^{n-1}.$$

This identity is the precise combinatorial place where the binary symmetry of the binomial expansion enters the argument.

Therefore, we may introduce a second positive real residual factor $q > 0$, obtained after extracting the total odd coefficient mass 2^{n-1} :

$$S_n(m, p) = 2^{n-1} q^n.$$

Equivalently,

$$q^n = \frac{1}{2^{n-1}} \sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{2i+1}.$$

This is not a claim that the weighted binomial core is equal to the pure sum of its coefficients. Rather, it is the definition of the second real residual factor q after the coefficient mass 2^{n-1} has been extracted.

Thus the same difference of powers also admits the representation

$$(m^n + p^n)^n - (m^n - p^n)^n = 2^n q^n. \quad (2.14)$$

Comparing (2.13a) and (2.14), we obtain

$$2n l^n = 2^n q^n.$$

After division by $2n > 0$, this gives the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}. \quad (2.15)$$

Logarithmic form of the compatibility relation. Since $l > 0$ and $q > 0$, we may take logarithms in (2.15). This gives

$$n \ln l = n \ln q + (n-1) \ln 2 - \ln n.$$

Equivalently,

$$\ln \frac{l}{q} = \ln 2 - \frac{\ln(2n)}{n}. \quad (2.16)$$

Thus the comparison of the two residual factors can be expressed by the logarithmic gap

$$\Delta_n := \ln \frac{l}{q} = \frac{1}{n} \ln \frac{2^{n-1}}{n}. \quad (2.17)$$

This quantity measures the logarithmic imbalance between the two residual factors. It may be interpreted heuristically as an information-type or entropy-type gap between the two normalizations of the same odd binomial core. However, in the strict mathematical argument it is simply a logarithmic reformulation of the exact algebraic relation (2.15). No physical entropy principle is used here as an independent proof.

Zero logarithmic gap and residual-scale compatibility. The equality of the two residual scales is equivalent to the vanishing of the logarithmic gap:

$$l^n = q^n \iff l = q \iff \Delta_n = 0,$$

where the equivalence $l^n = q^n \iff l = q$ is valid because $l > 0$ and $q > 0$.

By (2.17), the condition $\Delta_n = 0$ is equivalent to

$$\ln \frac{2^{n-1}}{n} = 0.$$

Therefore,

$$\frac{2^{n-1}}{n} = 1,$$

and hence

$$n = 2^{n-1}. \quad (2.18)$$

Conversely, if $n = 2^{n-1}$, then (2.17) gives

$$\Delta_n = 0,$$

and therefore $l = q$, equivalently $l^n = q^n$.

Thus, under the two real normalizations of the same odd binomial core, we have the exact equivalence

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1}. \quad (2.19)$$

The quadratic case. For $n = 2$, the relevant row of Pascal's triangle is

$$1, \quad 2, \quad 1.$$

There is only one odd-position binomial coefficient in the difference of the two symmetric powers, namely

$$C_2^1 = 2.$$

The linear coefficient extracted in the first normalization is

$$n = 2,$$

and the total odd coefficient mass extracted in the second normalization is

$$2^{2-1} = 2.$$

Thus the two extracted coefficient masses coincide:

$$n = 2^{n-1}.$$

Consequently,

$$\Delta_2 = 0,$$

and the two residual factors coincide:

$$l^2 = q^2.$$

This is exactly the quadratic, Pythagorean case.

Why higher exponents create a positive logarithmic gap. For $n = 3$, the odd part of the binomial expansion contains the coefficients

$$C_3^1 = 3, \quad C_3^3 = 1,$$

so the total odd coefficient mass is

$$2^{3-1} = 4.$$

The first normalization extracts 3, whereas the second normalization extracts 4. Hence

$$l^3 = q^3 \frac{4}{3},$$

and therefore

$$\Delta_3 = \frac{1}{3} \ln \frac{4}{3} > 0.$$

For $n = 4$, the odd coefficients are

$$C_4^1 = 4, \quad C_4^3 = 4,$$

so their total mass is

$$2^{4-1} = 8.$$

The first normalization extracts 4, while the second extracts 8. Hence

$$l^4 = 2q^4,$$

and therefore

$$\Delta_4 = \frac{1}{4} \ln 2 > 0.$$

In general, for every integer $n > 2$,

$$2^{n-1} > n.$$

Therefore, by (2.17),

$$\Delta_n = \frac{1}{n} \ln \frac{2^{n-1}}{n} > 0 \quad (n > 2). \quad (2.20)$$

Equivalently,

$$l^n > q^n \quad (n > 2).$$

Thus the binomial coefficient symmetry does not prove that $l^n = q^n$ for all exponents. It proves the sharper statement that the zero-gap condition

$$\Delta_n = 0$$

is compatible with the binomial coefficient structure only in the cases

$$n = 1 \quad \text{or} \quad n = 2.$$

Analysis of the equation $n = 2^{n-1}$.

The equation (2.18) is equivalent to

$$2^n = 2n.$$

It has exactly two positive integer solutions:

$$n = 1, \quad n = 2.$$

Indeed,

$$2^{1-1} = 1, \quad 2^{2-1} = 2.$$

For every integer $n > 2$, the exponential term 2^{n-1} strictly dominates the linear term n . Hence

$$2^{n-1} > n \quad \text{for all } n > 2.$$

Therefore, equation (2.18) is impossible for every exponent higher than the square.

Conclusion: coefficient symmetry as a logarithmic compatibility obstruction.

The binomial symmetry of Pascal's triangle does not merely motivate the second normalization. It gives a precise compatibility test for the equality of the two residual factors. Under the two normalizations

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n,$$

the logarithmic gap is

$$\Delta_n = \ln \frac{l}{q} = \frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

The zero-gap condition

$$\Delta_n = 0$$

is equivalent to

$$l^n = q^n,$$

and also equivalent to

$$n = 2^{n-1}.$$

Since this equation has only the positive integer solutions

$$n \in \{1, 2\},$$

the zero-gap condition is possible only in the linear and quadratic cases. For every $n > 2$, the logarithmic gap is strictly positive:

$$\Delta_n > 0.$$

Thus, in Version 16, the role of the binomial symmetry can be expressed in a logarithmic language: higher exponents create a strictly positive logarithmic imbalance between the two residual factors. The quadratic case is precisely the case where this imbalance vanishes.

This logarithmic formulation may be heuristically interpreted as an entropy-type separation of the two binomial scales. However, the strict mathematical content remains the exact equivalence

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1}.$$

Therefore, unless one proves independently that a hypothetical positive integer solution must force the zero-gap condition, the model remains conditional. If such a zero-gap principle is justified, then every exponent higher than the square is excluded.

6.) Verification (without considering the total multiplier r from (2.7), which has been reduced)

(a) Consider the case $n = 1$

$$\begin{aligned} z &= m + p \\ x &= m - p \\ y &= 2 \cdot [C_1^1(m^1)^{1-(2 \cdot 0+1)}(p^1)^{(2 \cdot 0+1)}] = 2[1 \cdot 1 \cdot p] = 2p \end{aligned}$$

that is, for the case $n = 1$, we have a solution in natural numbers x, y, z , if m and p - are positive integer numbers and $m > p$.

(b) Consider the case $n = 2$

$$\begin{aligned} z &= m^2 + p^2 \\ x &= m^2 - p^2 \\ y &= \sqrt{2} \cdot [C_1^2(m^2)^{2-(2 \cdot 0+1)}(p^2)^{(2 \cdot 0+1)}]^{1/2} = \sqrt{2} \cdot [2m^2p^2]^{1/2} = 2mp \end{aligned}$$

that is, for the case $n = 2$, we also have a solution in positive integer numbers x, y, z , if m and p are positive integer numbers and $m > p$, or according to [10], if $m > p$, and m, p are such non-integer numbers that combinations with them will give the positive integer numbers x, y, z , for example: $m = 3/\sqrt{2}$, $p = 1/\sqrt{2}$, in this case, we also get the classical Pythagorean triple:

$$\begin{aligned} x &= m^2 - p^2 = \left(\frac{3}{\sqrt{2}}\right)^2 - \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{9}{2} - \frac{1}{2} = \frac{8}{2} = 4 \\ y &= 2 \cdot m \cdot p = 2 \cdot \frac{3}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = 3 \\ z &= m^2 + p^2 = \left(\frac{3}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{9}{2} + \frac{1}{2} = 5 \end{aligned}$$

We have obtained the classical Pythagorean triple and can see that this formula works.

7). Let us now consider the more general case where r is not reduced to proceed to final formulas like (2.7). If r is a positive integer number, then all the above applies. But according to [10], r - can be a rational number. For example, let's say $r = 0.5$:

(a) Consider the case of $n = 1$, $m = 4$, and $p = 2$

$$\begin{aligned} x &= 0.5 \cdot (m - p) = 0.5 \cdot (4 - 2) = 1 \\ z &= 0.5 \cdot (m + p) = 0.5 \cdot (4 + 2) = 3 \\ y &= 0.5 \cdot (2 \cdot p) = 0.5 \cdot (2 \cdot 2) = 2 \end{aligned}$$

We see positive integer numbers. That is, m, p must be such that 7(a) is performed similarly. In this case, we can see that the formula 6(a) works.

(b) Consider the case of $n = 2$, $m = 2 \cdot \sqrt{2}$, $p = \sqrt{2}$,
 $x = 0.5 \cdot (m^2 - p^2) = 0.5 \cdot \left((2 \cdot \sqrt{2})^2 - (\sqrt{2})^2 \right) = 3$
 $z = 0.5 \cdot (m^2 + p^2) = 0.5 \cdot \left((2 \cdot \sqrt{2})^2 + (\sqrt{2})^2 \right) = 5$
 $y = 0.5 \cdot (2 \cdot m \cdot p) = 0.5 \cdot (2 \cdot (2 \cdot \sqrt{2}) \cdot (\sqrt{2})) = 4$

We see positive integer numbers. That is,, m , p must be such that 7(b) holds similarly. In this case, we can see that the formula 6(b) works.

This preserves the generality of solutions for $n = 2$, as seen in the Pythagorean triple example.

- 8). Therefore, within the zero-gap compatibility framework developed above, the only admissible positive integer exponents are

$$n = 1 \quad \text{and} \quad n = 2.$$

This is consistent with the classical linear and quadratic cases. The examples above show that for $n = 1$ and $n = 2$ the parametrization produces positive integer solutions in the cases considered, including the usual Pythagorean case for $n = 2$.

- 9). **under the zero-gap compatibility requirement, the equation $x^n + y^n = z^n$ can have compatible positive integer solutions only in the cases $n = 1$ and $n = 2$. For every $n > 2$, the logarithmic gap Δ_n is strictly positive, so the zero-gap condition is impossible.**

This completes the conditional algebraic argument based on the logarithmic zero-gap compatibility obstruction.

3 Remark and corollaries

REMARK. Note that the expression (2.8) can be simplified, namely

$$\begin{aligned} y &= \sqrt[n]{2} \left[\sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{(2i+1)} \right]^{1/n} = \\ &= \sqrt[n]{2} \left[\sum_{i=0}^k C_n^{(2i+1)} \frac{(m^n)^n}{(m^n)^{2i} m^n} (p^n)^{2i} p^n \right]^{1/n} = \\ &= \sqrt[n]{2} m^n \frac{1}{m} p \left[\sum_{i=0}^k C_n^{(2i+1)} \left(\frac{p}{m} \right)^{n2i} \right]^{1/n} = \\ &= \sqrt[n]{2} m^{n-1} p \left[\sum_{i=0}^k C_n^{(2i+1)} \left(\frac{p}{m} \right)^{2in} \right]^{1/n} \end{aligned} \tag{3.1}$$

with $k = (n - 1)/2$ if n is odd and $k = (n - 2)/2$ if n is even.

COROLLARY 1. Consider the case $m = p$, then from the expression (3.1) it can be derived that

$$\begin{aligned} x &= 0 \\ z &= m^n + m^n = 2m^n \\ y &= \sqrt[n]{2} m^{n-1} m \left[\sum_{i=0}^k C_n^{(2i+1)} \left(\frac{m}{m} \right)^{2in} \right]^{1/n} = \sqrt[n]{2} m^n \left[\sum_{i=0}^k C_n^{(2i+1)} \right]^{1/n} \end{aligned}$$

with $k = (n - 1)/2$ if n is odd and $k = (n - 2)/2$ if n is even.

It is obvious that $y = z$

$$\begin{aligned} \sqrt[n]{2} m^n \left[\sum_{i=0}^k C_n^{(2i+1)} \right]^{1/n} &= 2m^n \\ \sqrt[n]{2} \left[\sum_{i=0}^k C_n^{(2i+1)} \right]^{1/n} &= 2 \end{aligned} ,$$

whence

$$\sum_{i=0}^k C_n^{(2i+1)} = 2^{n-1} \quad (3.2)$$

with $k = (n - 1)/2$ if n is odd and $k = (n - 2)/2$ if n is even.

COROLLARY 2. *Based on (3.2), the sum of even combinations can be calculated. Consider Pascal's triangle [7]:*

$$\begin{array}{ccccccc} & & 1 & & 1 & & \\ & 1 & & 2 & & 1 & \\ & 1 & 3 & 3 & 1 & & \\ & 1 & 4 & 6 & 4 & 1 & \\ 1 & 5 & 10 & 10 & 5 & 1 & \end{array}$$

Similarly to the above, it is concluded that

$$\sum_{j=0}^s C_n^{2j} = 2^{n-1} \quad (3.3)$$

with $s = (n - 1)/2$ if n is odd and $s = n/2$ if n is even.

PROOF.

Expand

$$(m^n + p^n)^n + (m^n - p^n)^n =$$

into a binomial [8, 9]:

$$\begin{aligned} &= [(m^n)^n + C_n^1(m^n)^{n-1}p^n + C_n^2(m^n)^{n-2}(p^n)^2 + \dots + C_n^{n-1}m^n(p^n)^{n-1} + (p^n)^n] + \\ &+ [(m^n)^n - C_n^1(m^n)^{n-1}p^n + C_n^2(m^n)^{n-2}(p^n)^2 \pm \dots \pm C_n^{n-1}m^n(p^n)^{n-1} \pm (p^n)^n] = \\ &= 2C_n^0(m^n)^n + 2C_n^2(m^n)^{n-2}(p^n)^2 + \dots + 2C_n^k(m^n)^{n-k}(p^n)^k + \{2C_n^n(p^n)^n\} = \\ &= 2 \sum_{j=0}^s C_n^{2j} (m^n)^{n-2j} (p^n)^{2j} \end{aligned}$$

with $s = (n - 1)/2$ if n is odd and $s = n/2$ if n is even.

If $m = p$

$$(p^n + p^n)^n + (p^n - p^n)^n = 2 \sum_{j=0}^s C_n^{2j} (p^n)^{n-2j} (p^n)^{2j}$$

$$(2p^n)^n = 2 \sum_{j=0}^s C_n^{2j} \frac{(p^n)^n}{(p^n)^{2j}} (p^n)^{2j}$$

$$2^n (p^n)^n = 2 (p^n)^n \sum_{j=0}^s C_n^{2j}$$

$$2^{n-1} = \sum_{j=0}^s C_n^{2j}$$

with $s = (n - 1)/2$ if n is odd and $s = n/2$ if n is even.

Corollary 2 is proved.

COROLLARY 3. *Analysing (3.2) and (3.3), it can be concluded that*

$$\sum_{i=0}^k C_n^{(2i+1)} = \sum_{j=0}^s C_n^{2j} \quad (3.4)$$

with $k = (n-1)/2$, $s = (n-1)/2$ if n is odd and $k = (n-2)/2$, $s = n/2$ if n is even.
Why such borders?

Type of number	Condition	Last constant in equation
Odd	$k = 2((n-1)/2)+1 = n$	C_n^n
	$s = 2((n-1)/2) = n-1$	C_n^{n-1}
Even	$k = 2((n-2)/2)+1 = n-1$	C_n^{n-1}
	$s = 2(n/2) = n$	C_n^n

Table 1: Boundaries of binomial coefficients

Conclusion: the sum of even coefficients is equal to the sum of odd ones and is equal to 2^{n-1} , therefore from (3.4) we have

$$\sum_{r=0}^n C_n^r = \sum_{i=0}^k C_n^{(2i+1)} + \sum_{j=0}^s C_n^{2j} = 2 \cdot 2^{n-1} = 2^n \quad (3.5)$$

with $k = (n-1)/2$, $s = (n-1)/2$ if n is odd and $k = (n-2)/2$, $s = n/2$ if n is even.

4 Conclusion

The “difficulties” were for Fermat the lengthiness of the run of his deductions *put in writing*, since in the first half of the seventeenth century mathematical notation was still far from its present concise and diverse form, and many actions had to be written down *in words*. Besides, a purely mathematical challenge was that he had to operate with notions of binomial expansion and logarithmic comparison which were then still new, only emerging for systematic use and having to be learnt almost “on the fly”.

As mentioned in the introduction, the elementary algebraic part of the present reconstruction is accessible at the level of early university mathematics. This should be understood only as a comment on the elementary character of the reconstructed algebraic language, not as a comparison of mathematical strength with the modern proof of Fermat’s Last Theorem.

Andrew Wiles’s proof, completed with Richard Taylor, belongs to a very different modern framework involving modular forms, elliptic curves, Galois representations, and deformation theory. The present reconstruction does not compete with that proof. Rather, it explores whether a much more elementary algebraic line of thought could plausibly explain the kind of reasoning Fermat may have had in mind.

Fermat was, plausibly, “playing” with the new algebraic ideas of his time, decomposing powers of sums and differences and observing the special role of the binary structure. In the reconstruction proposed here, the decisive point is expressed not as an external normalization, but as a compatibility obstruction between two natural real normalizations of the same odd binomial core.

Indeed, the odd binomial core may be written in two ways:

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n.$$

Equivalently, for the full difference of powers, this gives

$$y^n = 2n l^n, \quad y^n = 2^n q^n.$$

Comparing these two representations yields the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Since $l > 0$ and $q > 0$, this relation can be written in logarithmic form. Define the logarithmic compatibility gap

$$\Delta_n := \ln \frac{l}{q} = \frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

This quantity measures the logarithmic imbalance between the two residual factors obtained from the two real normalizations of the same odd binomial core.

The zero-gap condition

$$\Delta_n = 0$$

is equivalent to the equality of the two residual scales:

$$l^n = q^n.$$

At the same time, by the formula for Δ_n , the zero-gap condition is also equivalent to

$$n = 2^{n-1}.$$

Thus the coefficient-symmetry compatibility obstruction of the present version may be written as the chain

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1}.$$

The elementary exponential-linear equation

$$n = 2^{n-1}$$

has positive integer solutions only for

$$n = 1 \quad \text{and} \quad n = 2.$$

For every integer $n > 2$, one has

$$2^{n-1} > n.$$

Consequently,

$$\Delta_n = \frac{1}{n} \ln \frac{2^{n-1}}{n} > 0 \quad (n > 2).$$

Therefore, for every exponent higher than the square, the logarithmic gap is strictly positive, and the zero-gap condition is impossible.

This is the sharpened role of the binomial symmetry in the present logarithmic formulation. It does not merely motivate a scale condition. It gives a precise compatibility test: the equality of the two residual scales, equivalently the vanishing of the logarithmic gap, is compatible with the binomial coefficient structure only in the cases $n = 1$ and $n = 2$. For every $n > 2$, the total odd binomial coefficient mass 2^{n-1} strictly exceeds the linear coefficient n , and the two residual factors necessarily separate.

The logarithmic gap

$$\Delta_n$$

may be interpreted heuristically as an entropy-type separation between the two binomial residual factors. In this interpretation, the quadratic case is the zero-gap case, while higher exponents create a strictly positive imbalance. However, this entropy-type language is only a heuristic interpretation. The strict mathematical content is the exact algebraic-logarithmic relation above. No physical entropy principle is used here as an independent proof.

Thus, if the internal two-power symmetry of a hypothetical positive integer solution is required to

preserve zero logarithmic gap between the two residual factors, equivalently to preserve

$$l^n = q^n,$$

then the equation

$$x^n + y^n = z^n$$

cannot have positive integer solutions for any $n > 2$.

This final step remains conditional in the precise sense stated throughout the paper. The Coq formalization checks the formally stated algebraic core of this chain: the two real factorizations of the same odd binomial core imply the exact relation between l and q , and the resulting exponential-linear condition forces

$$n \in \{1, 2\}.$$

The logarithmic formulation through Δ_n is mathematically equivalent to this algebraic relation and can be formalized as an additional explanatory layer. What Coq does not prove by itself is that every hypothetical positive integer solution of Fermat's equation must necessarily force the zero-gap condition

$$\Delta_n = 0.$$

This is the central mathematical point submitted for criticism.

He would first have had to introduce and explain these algebraic and logarithmic ideas in order to make his finding fully intelligible. One can imagine *how much room it would take to put down all the deliberations* that had led him to his discovery *on the margin of a book, without the symbolic notation available to a contemporary mathematician*.

Why Pierre Fermat did not write down all those ideas in a dedicated document is a separate historical question. It may turn out that he did write such a document, which was afterwards lost, or that it has survived to this day hidden in an archive, a library, or some unrealised private custody.

The author requests the mathematical community to look critically at the deliberations set forth above and to return their assessment, especially concerning the exact status of the zero-gap requirement and its possible connection with the internal two-power symmetry of a hypothetical positive integer solution.

This paper has been published as a preprint on the ResearchGate platform and on the OSF platform at the following links [11, 12].

The Coq formalization is available in the repository FLT-Coq on GitHub [14].

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A Appendix A. Proof of the Irrationality of $\sqrt[n]{2}$ for $n \geq 2$

A.1 Theorem: The number $\sqrt[n]{2}$ is irrational for any integer $n \geq 2$.

A.2 Proof:

We will prove by contradiction. Assume that $\sqrt[n]{2}$ is a rational number. This means it can be represented as an irreducible fraction $\frac{p}{q}$, where p and q are integers, $q \neq 0$, and p and q have no common divisors other than 1.

Thus, we have:

$$\sqrt[n]{2} = \frac{p}{q}$$

Raise both sides of the equation to the power of n :

$$(\sqrt[n]{2})^n = \left(\frac{p}{q}\right)^n$$

$$2 = \frac{p^n}{q^n}$$

Multiply both sides by q^n :

$$2q^n = p^n$$

From this equation, we see that p^n is an even number, since it is equal to $2q^n$. If p^n is even, then p must also be even (if p were odd, then p^n would also be odd).

Since p is even, we can write it as $p = 2k$, where k is some integer. Substitute this expression for p into the equation $2q^n = p^n$:

$$2q^n = (2k)^n$$

$$2q^n = 2^n k^n$$

Divide both sides of the equation by 2 (which is permissible since $n \geq 2$, and therefore 2^n is divisible by 2):

$$q^n = 2^{n-1} k^n$$

Since $n \geq 2$, then $n - 1 \geq 1$. From the last equation, we see that q^n is an even number, since it is equal to $2^{n-1} k^n$, where 2^{n-1} is an even factor. If q^n is even, then q must also be even.

Thus, we have reached the conclusion that both p and q are even numbers. This means they have a common divisor of 2. However, at the beginning of the proof, we assumed that the fraction $\frac{p}{q}$ was irreducible, meaning p and q have no common divisors other than 1.

The resulting contradiction indicates that our initial assumption about the rationality of $\sqrt[n]{2}$ is incorrect.

A.3 Conclusion:

Therefore, the number $\sqrt[n]{2}$ is irrational for any integer $n \geq 2$.

B Appendix B. Proof that the limit of the function $f(n) = \sqrt[n]{2 \cdot n}$ is 1

B.1 Problem Statement

Prove that the limit of the function $f(n) = \sqrt[n]{2 \cdot n}$ as n approaches infinity is equal to 1.

B.2 Solution

Consider the function $f(n) = \sqrt[n]{2 \cdot n}$. Our goal is to find the limit of this function as $n \rightarrow \infty$. Let's rewrite the function in the form of a power:

$$f(n) = (2 \cdot n)^{\frac{1}{n}}$$

To find the limit of this function, let's consider the natural logarithm of $f(n)$:

$$\ln(f(n)) = \ln\left((2 \cdot n)^{\frac{1}{n}}\right)$$

Using the properties of logarithms, we get:

$$\ln(f(n)) = \frac{1}{n} \ln(2 \cdot n)$$

Separate the logarithm of the product into the sum of logarithms:

$$\ln(f(n)) = \frac{\ln(2) + \ln(n)}{n}$$

Now, let's find the limit of $\ln(f(n))$ as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \ln(f(n)) = \lim_{n \rightarrow \infty} \frac{\ln(2) + \ln(n)}{n}$$

This limit has the indeterminate form $\frac{\infty}{\infty}$, so we can apply L'Hôpital's Rule. Take the derivatives of the numerator and denominator with respect to n :

$$\frac{d}{dn}(\ln(2) + \ln(n)) = \frac{1}{n}$$

$$\frac{d}{dn}(n) = 1$$

Thus, the limit becomes:

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{1} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

So, we have found that the limit of the natural logarithm of the function is 0:

$$\lim_{n \rightarrow \infty} \ln(f(n)) = 0$$

Now, to find the limit of the original function $f(n)$, we use the continuity of the exponential function:

$$\lim_{n \rightarrow \infty} f(n) = \lim_{n \rightarrow \infty} e^{\ln(f(n))} = e^{\lim_{n \rightarrow \infty} \ln(f(n))} = e^0 = 1$$

B.3 Conclusion

Therefore, we have proven that the limit of the function $f(n) = \sqrt[n]{2 \cdot n}$ as n approaches infinity is equal to 1.

C Appendix C. A short proof of monotonicity

It is well known that $\sqrt[n]{2 \cdot n} \rightarrow 1$ as $n \rightarrow \infty$ (see Appendix B). However, in this appendix we focus on showing that $(2 \cdot n)^{1/n}$ is strictly decreasing for $n \geq 3$. Below is one way to prove this, using the logarithmic derivative:

C.1 Define the function.

Let

$$f(n) = (2 \cdot n)^{\frac{1}{n}}.$$

It is convenient to work with its natural logarithm:

$$\ln(f(n)) = \ln((2 \cdot n)^{\frac{1}{n}}) = \frac{1}{n} \ln(2 \cdot n).$$

Denote

$$g(n) = \ln(f(n)) = \frac{\ln(2 \cdot n)}{n} = \frac{\ln(2) + \ln(n)}{n}.$$

C.2 Compute the derivative.

Treating n as a real variable $n > 0$, we have

$$g'(n) = \frac{d}{dn} \left(\frac{\ln(2 \cdot n)}{n} \right).$$

Using the quotient rule,

$$g'(n) = \frac{\frac{d}{dn}[\ln(2 \cdot n)] \cdot n - \ln(2 \cdot n) \cdot 1}{n^2} = \frac{\frac{1}{n} \cdot n - \ln(2 \cdot n)}{n^2} = \frac{1 - \ln(2 \cdot n)}{n^2}.$$

(Note that $\frac{d}{dn}[\ln(2 \cdot n)] = \frac{1}{2 \cdot n} \cdot 2 = \frac{1}{n}$.)

C.3 Sign of the derivative.

We want to see where $g'(n) < 0$:

$$g'(n) < 0 \iff 1 - \ln(2 \cdot n) < 0 \iff \ln(2 \cdot n) > 1 \iff 2 \cdot n > e \iff n > \frac{e}{2}.$$

Since $e \approx 2.718$, the inequality $n > e/2$ is certainly true for all integer $n \geq 2$, and hence strictly for $n \geq 3$. Therefore, for $n \geq 3$, $g(n)$ is strictly decreasing.

C.4 Implication for $f(n)$.

Since $f(n) = \exp(g(n))$, and $\exp(\cdot)$ is a strictly increasing function in its argument, $f(n)$ decreases whenever $g(n)$ decreases. Hence, $f(n)$ is indeed strictly decreasing for $n \geq 3$.

Thus, only for $n = 1$ and $n = 2$ does $f(n)$ attain the maximum value 2, and for all $n \geq 3$, it strictly decreases (which is consistent with the limit $\lim_{n \rightarrow \infty} (2 \cdot n)^{1/n} = 1$).

C.5 Conclusion

Hence, we have demonstrated that $f(n)$ is strictly decreasing for $n \geq 3$, either by analyzing the derivative of $\ln(f(n))$ (as shown above) or by comparing $f(n+1)$ and $f(n)$. This completes the proof of the monotonicity of $f(n)$.

D Appendix D: Logarithmic Compatibility Gap and Heuristic Visualization of Critical Points

D.1 Introduction

The purpose of this appendix is to provide a geometric and logarithmic visualization of the coefficient-symmetry compatibility obstruction discussed in the main text. The appendix is not intended as an independent proof of Fermat's Last Theorem. It is only an auxiliary illustration of the same conditional structure.

In the main text, the decisive algebraic comparison comes from two natural real normalizations of the same odd binomial core:

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n.$$

Equivalently, for the full symmetric difference of powers,

$$y^n = 2n l^n, \quad y^n = 2^n q^n.$$

Here $l > 0$ and $q > 0$ are positive real residual factors. The symbol q denotes a residual factor and must not be confused with a modulus in modular arithmetic.

Comparing the two normalizations gives the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Since $l > 0$ and $q > 0$, this relation can be written in logarithmic form:

$$n \ln l = n \ln q + (n - 1) \ln 2 - \ln n.$$

After division by n , this becomes

$$\ln \frac{l}{q} = \frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

We therefore introduce the logarithmic compatibility gap

$$\Delta_n := \ln \frac{l}{q} = \frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

The zero-gap condition

$$\Delta_n = 0$$

is equivalent to

$$l = q,$$

and, since $l, q > 0$, also to

$$l^n = q^n.$$

By the formula for Δ_n , the same zero-gap condition is equivalent to

$$n = 2^{n-1}.$$

Thus the central compatibility chain is

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1}.$$

The equation

$$n = 2^{n-1}$$

has exactly two positive integer solutions:

$$n = 1, \quad n = 2.$$

For every integer $n > 2$, one has

$$2^{n-1} > n,$$

and therefore

$$\Delta_n = \frac{1}{n} \ln \frac{2^{n-1}}{n} > 0.$$

Consequently, the logarithmic gap vanishes only in the linear and quadratic cases and becomes strictly positive for all exponents higher than the square.

This gap may be interpreted heuristically as an entropy-type separation between the two residual factors obtained from the two real normalizations of the same odd binomial core. However, in the strict mathematical argument it is only a logarithmic reformulation of the exact algebraic relation above. No physical entropy principle is used here as an independent proof.

D.2 Auxiliary Function for Geometric Visualization

To visualize the special role of the binary/quadratic point, we consider the auxiliary continuous function

$$F(P, Q) = Q - \sqrt[n]{QP},$$

where $P > 0$ and $Q > 0$ are auxiliary real variables.

In this appendix, the symbol P is used as a continuous variable comparable with the discrete exponent n . It should not be confused with the parameter p in the main parametrization

$$z = m^n + p^n, \quad x = m^n - p^n.$$

Likewise, Q is an auxiliary positive real visualization parameter. It should not be confused with the residual factor q in the main algebraic text, nor with a modulus in modular arithmetic.

The point

$$P = 2, \quad Q = 2$$

is used only as a geometric marker of the binary/quadratic case. It is not an additional axiom, not a replacement for the algebraic compatibility relation, and not a graphical proof of Fermat's Last Theorem.

The strict algebraic statement remains

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1}.$$

The auxiliary function $F(P, Q)$ is used only to provide a visual intuition for why the binary/quadratic value is distinguished.

D.3 Combined Graph of the Auxiliary Function

The following graph, Fig. 1, illustrates the functions

$$Q - \sqrt[p]{QP}$$

for several fixed values of Q , together with their corresponding discrete values at integer points.

For example, one may compare:

- $1.5 - \sqrt[p]{1.5P}$ and the corresponding discrete values,
- $2 - \sqrt[p]{2P}$ and the corresponding discrete values,
- $3 - \sqrt[p]{3P}$ and the corresponding discrete values.

These curves should be read as a visual comparison between continuous values of P and discrete exponent values n . The case $Q = 2$ is singled out only because it corresponds to the binary value appearing in the equation

$$n = 2^{n-1} \quad \Longleftrightarrow \quad 2^n = 2n.$$

In the main text, this relation is not derived from the graph. It follows from the comparison of the two binomial normalizations:

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n,$$

together with the zero-gap condition

$$\Delta_n = 0,$$

or equivalently

$$l^n = q^n.$$

Thus the graph does not prove Fermat's Last Theorem and does not prove the algebraic equivalence. It only visualizes the same quadratic compatibility point.

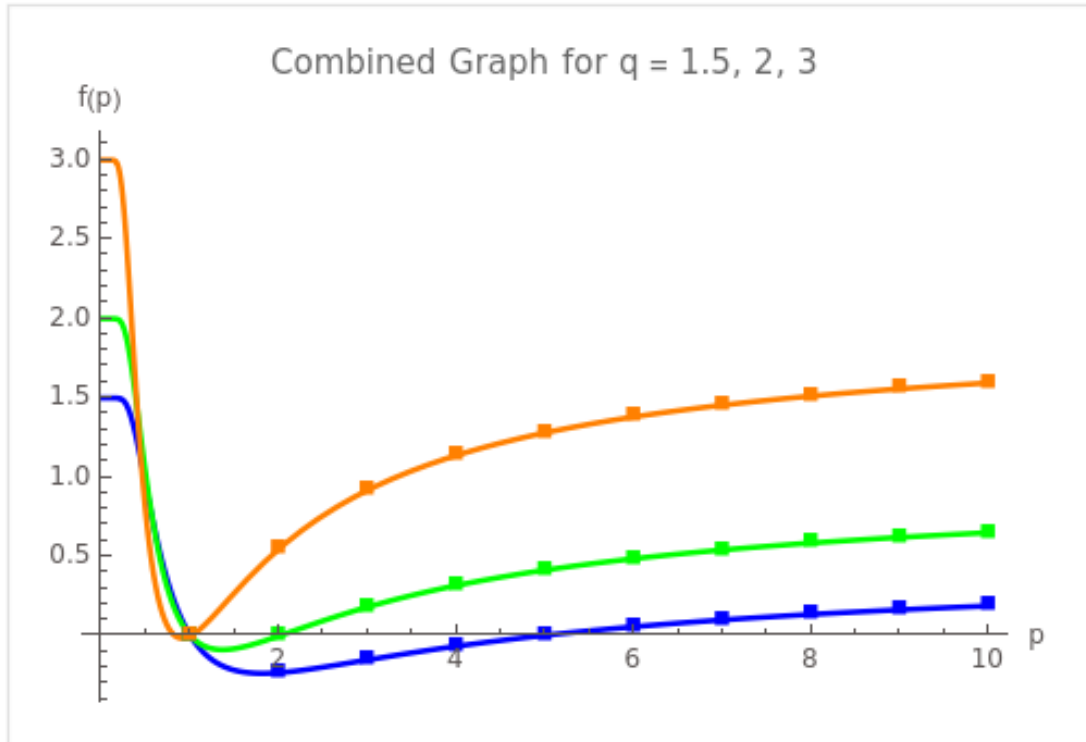


Figure 1: Combined graph of $F(P, Q) = Q - \sqrt[p]{QP}$ for several fixed values of Q . Lines represent continuous P , and markers represent discrete values. The value $Q = 2$ marks the binary/quadratic alignment only as a geometric visualization.

D.4 Surface Plot of the Auxiliary Function

The three-dimensional surface plot in Fig. 2 illustrates the behavior of

$$F(P, Q) = Q - \sqrt[p]{QP}$$

for continuous values $P > 0$ and $Q > 0$.

This visualization is meant to support the geometric intuition behind the binary value $Q = 2$. The algebraic argument itself rests on the comparison

$$y^n = 2n l^n, \quad y^n = 2^n q^n,$$

which gives

$$l^n = q^n \frac{2^{n-1}}{n}.$$

After logarithmization, the same comparison gives the gap

$$\Delta_n = \ln \frac{l}{q} = \frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

Therefore, the surface plot is only a heuristic visual aid. It does not prove Fermat's Last Theorem and does not replace the algebraic logarithmic compatibility obstruction.

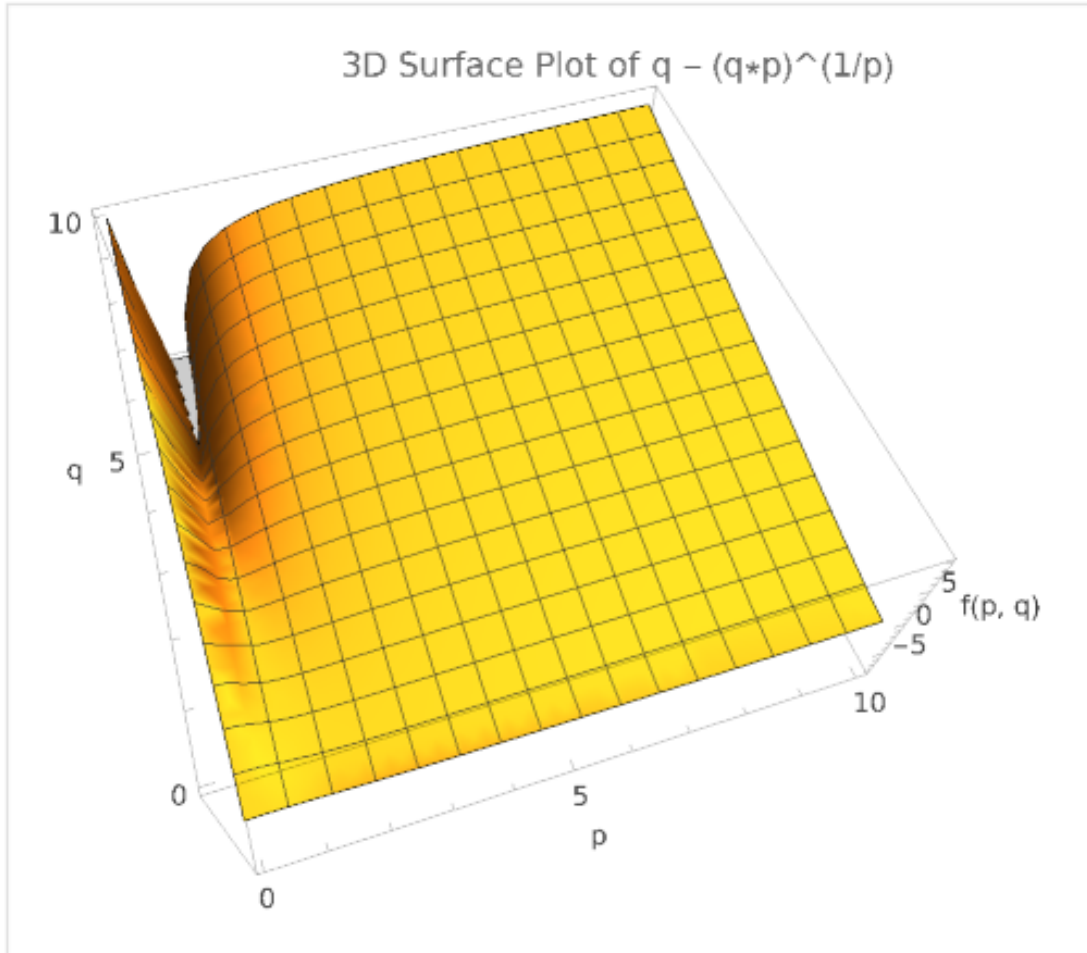


Figure 2: Surface plot of $F(P, Q) = Q - \sqrt[p]{QP}$ for $P > 0$ and $Q > 0$. The point $P = 2$, $Q = 2$ is interpreted only as a geometric marker of the quadratic alignment.

D.5 Corrected Interpretation of Critical Points

The numerical and graphical analysis of the auxiliary function suggests that the point

$$P = 2, \quad Q = 2$$

is visually distinguished. However, this statement must be understood carefully.

The strict algebraic obstruction in the main text is not obtained from a critical point of $F(P, Q)$. It is obtained from the exact comparison of the two normalizations of the odd binomial core. The logarithmic form of that comparison is

$$\Delta_n = \frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

The zero-gap condition

$$\Delta_n = 0$$

holds exactly for

$$n = 1, \quad n = 2,$$

and for every integer $n > 2$ one has

$$\Delta_n > 0.$$

Thus, any graphical or differential feature of the auxiliary function should be read only as a heuristic consistency check. It cannot replace the algebraic argument and cannot by itself justify the zero-gap condition for a hypothetical integer solution.

D.6 Conclusion of Appendix D

The function

$$F(P, Q) = Q - \sqrt[n]{QP}$$

provides a useful geometric illustration of the binary/quadratic point. It supports the intuition that the value 2 is special in the comparison between a linear quantity and a binary exponential quantity.

However, the rigorous part of the article remains the algebraic-logarithmic compatibility chain

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1} \implies n \in \{1, 2\}.$$

For all exponents

$$n > 2,$$

the logarithmic gap is strictly positive:

$$\Delta_n > 0.$$

Therefore, the zero-gap condition is impossible for every exponent higher than the square.

This appendix should be read only as a geometric and heuristic visualization of that obstruction, not as an independent proof.

E Appendix E: Geometric and Logarithmic Motivation for the Coefficient-Symmetry Compatibility Obstruction

E.1 Purpose of This Appendix

This appendix further investigates the connection between the auxiliary function

$$F(P, Q) = Q - \sqrt[p]{QP}$$

and the logarithmic compatibility gap associated with the two normalizations of the odd binomial core.

The purpose of this appendix is not to provide a standalone geometric proof of Fermat's Last Theorem. Its purpose is to explain how the geometric picture is compatible with the algebraic-logarithmic obstruction developed in the main article.

In the coefficient-symmetry compatibility scheme, the decisive step is not a graphical property of $F(P, Q)$. The decisive step is the comparison of two real normalizations of the same odd binomial core:

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n.$$

Equivalently,

$$y^n = 2n l^n, \quad y^n = 2^n q^n.$$

Their comparison gives

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Taking logarithms gives

$$\Delta_n = \ln \frac{l}{q} = \frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

The zero-gap condition

$$\Delta_n = 0$$

is equivalent to

$$l^n = q^n$$

and also to

$$n = 2^{n-1}.$$

Thus the final obstruction is the elementary exponential-linear equation

$$n = 2^{n-1},$$

not a graphical property of $F(P, Q)$.

The function $F(P, Q)$ gives a visual way to see why the point

$$P = 2, \quad Q = 2$$

is geometrically distinguished. In this sense the function supports the geometric intuition behind the coefficient-symmetry compatibility obstruction, but it does not prove Fermat's Last Theorem and does not replace the algebraic comparison.

E.2 Analysis of the Second Derivative

The function

$$F(P, Q) = Q - \sqrt[p]{QP}$$

allows us to study changes in curvature with respect to the continuous variable P . The second derivative with respect to P is denoted by

$$F_P''(P, Q) = \frac{\partial^2}{\partial P^2} \left(Q - \sqrt[P]{QP} \right).$$

Writing

$$G(P, Q) = (QP)^{1/P},$$

we have

$$F(P, Q) = Q - G(P, Q).$$

A direct differentiation gives

$$F_P''(P, Q) = -G(P, Q) \left[\frac{(1 - \ln(QP))^2}{P^4} + \frac{2 \ln(QP) - 3}{P^3} \right].$$

Equivalently, the inflection condition is

$$(1 - \ln(QP))^2 + P(2 \ln(QP) - 3) = 0.$$

This formula is useful for numerical and geometric visualization, but it is not used in the proof of the algebraic compatibility obstruction. In particular, the statement that the point $P = 2$, $Q = 2$ is special should not be confused with a claim that the whole proof follows from the vanishing of this second derivative.

At $P = 2$, $Q = 2$, the point is visually and conceptually related to the quadratic alignment, but the exact obstruction in the article is still the zero-gap condition

$$\Delta_n = 0,$$

not the differential equation

$$F_P''(P, Q) = 0.$$

E.3 Surface Plot of the Second Derivative

The following surface plot, Fig. 3, visualizes the second derivative

$$F_P''(P, Q)$$

of the auxiliary function

$$F(P, Q) = Q - \sqrt[P]{QP}.$$

This graph is useful because it shows how the curvature of the auxiliary function changes when both variables P and Q vary. The colored curves on the surface may be read as visual sections of the same curvature landscape. In particular, they help to illustrate why the region near the binary/quadratic alignment

$$P = 2, \quad Q = 2$$

is visually distinguished.

However, the graph must be interpreted carefully. It does not prove Fermat's Last Theorem, and it does not prove the zero-gap condition

$$\Delta_n = 0.$$

It only visualizes the behavior of the curvature of the auxiliary function $F(P, Q)$. The strict algebraic-logarithmic obstruction remains

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1}.$$

Thus, the surface of the second derivative should be regarded as a geometric consistency illustration accompanying the logarithmic obstruction, not as an independent mathematical proof.

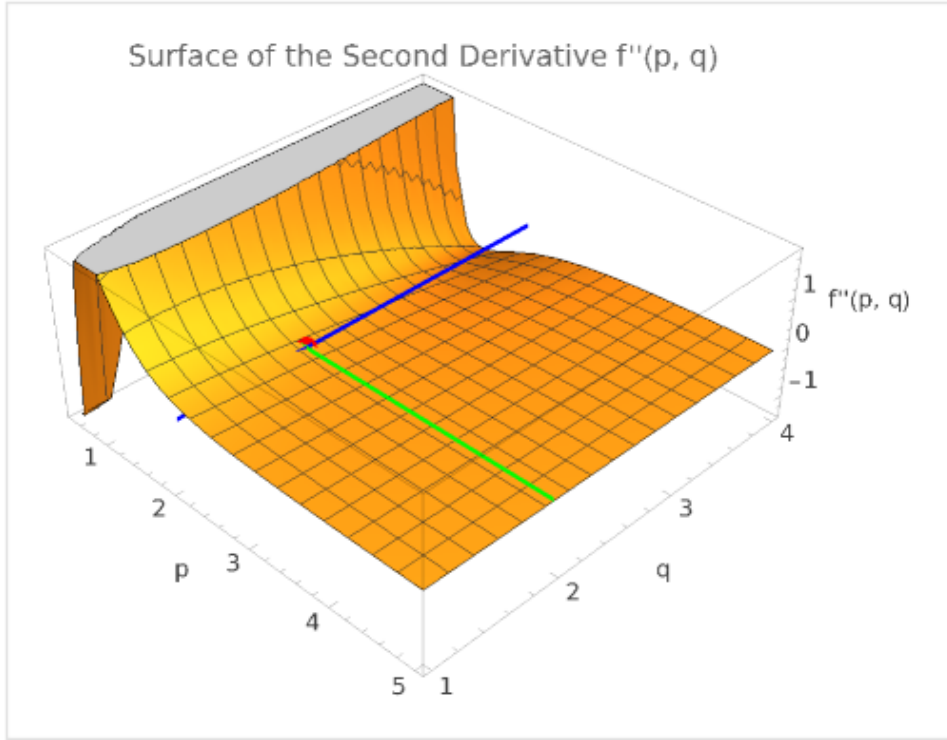


Figure 3: Surface of the second derivative $F''_P(P, Q)$ for the auxiliary function $F(P, Q) = Q - \sqrt[n]{QP}$. The graph visualizes the curvature landscape of the auxiliary function. It is used only as a heuristic geometric illustration and not as a proof of the zero-gap condition or of Fermat's Last Theorem.

E.4 Numerical Values of the Second Derivative

For reference, the following table gives numerical values of

$$F''_P(P, Q)$$

for several values of P and Q . The values are rounded and serve only as a numerical illustration.

P	$Q = 1.5$	$Q = 2.0$	$Q = 2.5$	$Q = 3.0$	$Q = 4.0$
1	2.7534	3.0391	2.9010	2.3792	0.3128
2	0.1728	0.0382	-0.1131	-0.2746	-0.6157
3	-0.0057	-0.0533	-0.0996	-0.1442	-0.2294
4	-0.0181	-0.0381	-0.0564	-0.0733	-0.1041
5	-0.0148	-0.0247	-0.0334	-0.0412	-0.0552

Table 2: Numerical values of the second derivative $F''_P(P, Q)$ for various P and Q . These values are illustrative only and are not used as a proof of the main theorem.

The table and the surface plot should be read together. The table gives selected numerical sections of the second derivative, while the surface plot gives a global geometric picture of the same curvature behavior. Neither of them is used as a proof of the algebraic compatibility obstruction.

In particular, the algebraic compatibility equivalence

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1}$$

does not come from the table and does not come from the graph. It comes from the comparison of the two real normalizations of the same odd binomial core.

E.5 The Logarithmic Gap as the Main Diagnostic

The main diagnostic quantity of the present formulation is not the second derivative of the auxiliary function but the logarithmic gap

$$\Delta_n = \frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

This quantity has the following properties:

- For $n = 1$,

$$\Delta_1 = 0.$$

- For $n = 2$,

$$\Delta_2 = 0.$$

- For every integer $n > 2$,

$$\Delta_n > 0.$$

Thus the quadratic case is exactly the last positive integer exponent for which the logarithmic gap vanishes. For higher exponents, the total odd coefficient mass 2^{n-1} strictly exceeds the linear coefficient n , and the residual factors separate logarithmically.

This can be written as

$$2^{n-1} > n \quad (n > 2),$$

and therefore

$$\frac{2^{n-1}}{n} > 1,$$

so

$$\Delta_n = \frac{1}{n} \ln \frac{2^{n-1}}{n} > 0.$$

Heuristically, this positive value may be described as an entropy-type separation between the two residual factors. However, this language is interpretative. The strict mathematical content is the positivity of the logarithmic gap.

E.6 Relation to Fermat's Equation

The heuristic geometric picture can be summarized as follows:

1. The point $P = 2$, $Q = 2$ is visually associated with the quadratic case.
2. The quadratic case is the case where Pythagorean triples exist.
3. In the algebraic-logarithmic formulation, the quadratic case is also the last positive integer exponent for which

$$\Delta_n = 0.$$

4. For every exponent $n > 2$, the logarithmic gap is strictly positive:

$$\Delta_n > 0.$$

5. Hence the zero-gap condition is incompatible with every exponent higher than the square.

This is consistent with the coefficient-symmetry compatibility obstruction:

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1} \implies n \in \{1, 2\}.$$

However, this appendix does not prove that every hypothetical positive integer solution of Fermat's equation must force

$$\Delta_n = 0.$$

That point remains the conditional structural requirement of the reconstruction.

E.7 Final Conclusion

The function

$$F(P, Q) = Q - \sqrt[P]{QP}$$

and its derivatives provide a useful geometric visualization of the special role of the binary/quadratic point

$$P = 2, \quad Q = 2.$$

The surface plot of the second derivative

$$F_P''(P, Q)$$

adds a further geometric illustration: it shows how the curvature of the auxiliary function changes over the (P, Q) -plane and gives a visual background for the numerical values listed in Table 2.

However, the strict mathematical obstruction developed in the main text is not a graphical or differential fact about this auxiliary function.

The strict obstruction is the algebraic-logarithmic chain

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n,$$

which gives

$$l^n = q^n \frac{2^{n-1}}{n}.$$

After logarithmization this becomes

$$\Delta_n = \ln \frac{l}{q} = \frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

The zero-gap condition

$$\Delta_n = 0$$

is equivalent to

$$l^n = q^n$$

and also to

$$n = 2^{n-1}.$$

Since this equation has only the positive integer solutions

$$n \in \{1, 2\},$$

the zero-gap condition is possible only in the linear and quadratic cases.

For every integer

$$n > 2,$$

the logarithmic gap is strictly positive:

$$\Delta_n > 0.$$

Therefore, if the internal two-power symmetry of a hypothetical positive integer solution is required to preserve zero logarithmic gap between the two residual factors, then every exponent higher than the square is excluded.

This conclusion is conditional. The geometric and logarithmic language clarifies the obstruction, and the surface of $F_P''(P, Q)$ helps visualize the curvature of the auxiliary function, but neither the graph nor the second derivative proves the zero-gap requirement for an arbitrary hypothetical integer solution of Fermat's equation.